Boolean Networks in Life Sciences
Exercise Sheet 3: Transition Systems

Wednesday 4th November, 2026

Exercise 1 *Construct the transition system of the following Boolean networks, for all of synchronous, fully asynchronous and generalised asynchronous semantics.*

1. $f_1(\mathbf{x}) = f_2(\mathbf{x}) = \mathbf{x}_1 \Leftrightarrow \mathbf{x}_2$;
2. $f_1(\mathbf{x}) = \neg\mathbf{x}_3$, $f_2(\mathbf{x}) = \neg\mathbf{x}_1$, $f_3(\mathbf{x}) = \neg\mathbf{x}_2$;
3. $f_1(\mathbf{x}) = (\mathbf{x}_1 \Leftrightarrow \mathbf{x}_2) \vee (\mathbf{x}_2 \oplus \mathbf{x}_3)$, $f_2(\mathbf{x}) = (\neg\mathbf{x}_1 \wedge \mathbf{x}_2) \vee (\mathbf{x}_1 \oplus \mathbf{x}_3)$, $f_3(\mathbf{x}) = \neg\mathbf{x}_2 \vee (\mathbf{x}_1 \Leftrightarrow \mathbf{x}_3)$;

Exercise 2 *For all the transition systems from Exercise 1, and for each configuration $\mathbf{x} \in \mathbb{B}^2$, resp. $\mathbf{x} \in \mathbb{B}^3$, obtain the smallest trap space $[\mathbf{x}]_{\mathcal{T}}$ containing $\mathbf{x}$ and compare them across the different semantics.*

Exercise 3 *Find all attractors of the Boolean networks from Exercise 1 for all of synchronous, fully asynchronous and generalised asynchronous semantics. Compare the discovered attractors against minimal trap spaces.*

Exercise 4 *Characterise the weak and strong basins of attraction of all attractors discovered in Exercise 3.*